

Mingyang (Yang) Liu

Visiting Researcher, Imperial Centre of Excellence in Quantitative Finance, Imperial College London

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On the 2026–27 academic job market

Research Fields

Empirical asset pricing (cross-sectional return prediction, characteristic-based factor models); machine learning and big data in finance (regularised and latent-factor estimators on large characteristic panels); financial econometrics (identification and inference in high-dimensional factor models); portfolio choice and risk management.

Positions

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| Jun 2026–present | <p>Visiting Researcher, Imperial Centre of Excellence in Quantitative Finance, Department of Finance, Imperial Business School, Imperial College London</p> <p>Host: Prof. Robert Kosowski, Co-Director of the Centre</p> <p>Research agenda for the visit: economically constrained machine learning for cross-sectional asset pricing and systematic investment.</p> <ul style="list-style-type: none">• Machine learning and high-dimensional data: extending the No-Arbitrage IPCA framework of the thesis to regime-switching factor models in which risk premia vary with macro conditions; combining penalised estimation with economically constrained representation learning to separate alpha signals that survive those constraints from signals driven by weak factors or data snooping.• Research and practice: using the quantitative platform of the Imperial Student Investment Fund to test research signals as implementable strategies, net of transaction costs and within turnover and risk limits. |
| Aug 2026–present | <p>AI and Machine-Learning Adviser (part-time), Cost of Capital team, Infrastructure Advisory, KPMG LLP (UK)</p> <ul style="list-style-type: none">• Academic advice on machine-learning methods and on the use of asset-pricing models, such as the conditional CAPM and the betting-against-beta (BAB) factor, in independent estimates of the allowed cost of equity for regulated UK utilities (water and energy networks). |

Education

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| Aug 2019–Aug 2026 | <p>Imperial College London, Imperial Business School, Department of Finance, London</p> <p>PhD in Finance, conferred August 2026</p> <p>Supervisors: Prof. Paolo Zaffaroni (main) and Prof. Pasquale Della Corte</p> <p>Thesis: No-Arbitrage IPCA: A Framework for Cross-Sectional Asset Pricing. Examined March 2026.</p> <p>MRes in Finance, with Distinction, 2020</p> |
| Sep 2017–Feb 2019 | <p>Columbia University, Graduate School of Arts and Sciences, New York</p> <p>M.A. in Economics (Financial Econometrics; STEM-designated programme)</p> <p>Adviser: Prof. Jushan Bai. Master's thesis: Diffusion Index Forecasting: A Robust Estimator for Common Shock</p> |
| Sep 2014–Apr 2017 | <p>University of Michigan, Ann Arbor</p> <p>B.S. in Economics and Mathematics</p> <p>Track: Mathematics of Finance and Risk Management</p> |

Research

Job Market Paper

Characteristic Libraries and Portfolio Decisions

Working paper, September 2026

Can a library change that adds no information alter a tuned investment rule? On 153 U.S. characteristics, theme balancing moves target holdings by 19–28% of average gross exposure across four ridge-type procedures and shifts theme attribution, with cost-dependent payoffs; duplicated characteristics act similarly. The 39-to-153 expansion has small, imprecise predictive effects under the evaluated procedures.

PDF

Working Papers

Characteristic-Space Metrics in Factor Models: Identification and Inference

SSRN preprint, September 2026. DOI: 10.2139/ssrn.7445120

Factor models with identical fit can attribute returns differently to latent factors, observed factors and an intercept. The inner product chosen for characteristic space selects this split; the paper formulates the estimand as a loading class modulo rotations and derives chart-invariant inference carrying the Gram metric's estimation error, and drift-robust intervals near an exposure bound.

PDF · Online supplement · SSRN

A Geometric Framework for Identification in Characteristic-Based Factor Models

SSRN preprint, March 2026 (preliminary version). DOI: 10.2139/ssrn.7013178. Grows out of the PhD thesis; its inference theory is developed further in Characteristic-Space Metrics.

Which directions in characteristic space are economically distinguishable? The paper takes the metric from the long-run cross-sectional distribution of firm characteristics, states identification and no-arbitrage restrictions as intrinsic geometric properties, and derives No-Arbitrage IPCA (NA-IPCA), which nests IPCA and admits rotation-invariant inference. In geometry-calibrated simulations, Euclidean representations become unstable in anisotropic characteristic spaces.

PDF · SSRN

Characteristic Geometry and Portfolio Choice

Working paper, September 2026. Slides available; manuscript in preparation.

Why do factor models with nearly identical forecasts imply different portfolios? On 132 U.S. characteristics, a fixed factor-variance ridge penalises exposures differently across loading representations; removing only that ridge shrinks the largest relative weight gap to about 0.01%. Within one NA-IPCA fit, using the intercept raises gross performance; the post-2007 mean-return gain is imprecisely estimated.

Slides (PDF)

Interpreting Estimated Pricing Errors: Evidence from Characteristic-Based Return Forecasts

Working paper, September 2026

What does a smaller estimated pricing error tell a model user? Using 153 U.S. characteristics, the paper separates changes in fitted expected returns from changes in factor loading spaces: the gap in fitted characteristic-intercept magnitudes between equal- and value-weighted training mainly reflects different estimated means, and residual diagnostics add little to forecast selection (exploratory evidence).

PDF

PhD Thesis

No-Arbitrage IPCA: A Framework for Cross-Sectional Asset Pricing

PhD thesis, Imperial College London; degree conferred August 2026. Thesis available on request.

Extends IPCA, via the empirical characteristic metric, to nested QZ-IPCA and NA-IPCA models separating latent, observable (Fama–French five factors plus momentum) and residual-pricing components under a no-arbitrage bound; gives identification results, an asymptotic linear expansion and projected alternating least squares; simulations and a Jensen–Kelly–Pedersen U.S. panel show that similar fit can hide different pricing directions.

Research Experience

Aug 2019–Aug 2026

Doctoral research, Department of Finance, Imperial College London (asset pricing, financial econometrics, machine learning in finance)

- Models and geometry: characteristic-based factor models with an explicit characteristic-space geometry, using the empirical Gram metric of firm characteristics as the identification convention; the extension of IPCA to the nested QZ-IPCA and NA-IPCA models developed in the thesis.

- Identification and inference: rotation-invariant identification (loading classes modulo latent rotations); asymptotic linear representations that carry an estimated metric's error into standard errors; drift-robust confidence intervals near an exposure bound; projected alternating least-squares estimation; geometry-calibrated Monte Carlo designs.
- Portfolio experiments and trading costs: rolling out-of-sample portfolio construction (120-month windows, U.S. stocks 1973–2024) comparing models under a common risk adjustment; information-preserving library interventions (exact copies, theme balancing) across four tuned ridge-type procedures, traced to theme attribution, stock-level ledgers and returns under transaction-cost scenarios; a separate 39-to-153 characteristic expansion evaluated for stock-return prediction; forecast-evaluation designs that cross expected-return estimates with factor spaces.
- Data and tools: the Jensen–Kelly–Pedersen library of 153 U.S. stock characteristics (jkpfactors.com; WRDS contrib.global_factor, built from CRSP/Compustat), 1962–2025, with point-in-time formation alignment; a CRSP/Compustat panel built following the Kelly–Pruitt–Su replication dataset (36 characteristics plus a constant, 12,813 firms, 1964–2014); CRSP daily volume for liquidity; NBER recession dates. Author of naipca, a Python toolkit implementing IPCA, QZ-IPCA and NA-IPCA with a companion metric-diagnostics package (available on request).

May 2018–May 2019 **Research Assistant (macro forecasting)**, Department of Economics, Columbia University

- Diffusion-index and factor models that extract common inflation and interest-rate factors from large mixed-frequency macro data; robust PCA and stable principal component pursuit for missing data, revisions, outliers and structural breaks.

Jun 2016–Jan 2017 **Research Assistant (derivatives pricing)**, Department of Mathematics, University of Michigan

- Real-option valuation of a gold mine, using the Lihir Gold Limited IPO as a case: estimated gold-price volatility and the present value of total costs, took strike parameters from the prospectus, and priced the option with a binomial tree.

Teaching

2023, 2024 **Graduate Teaching Assistant**, Imperial Business School
Financial Statistics, MSc Risk Management & Financial Engineering. Student rating 4.8 out of 5.
Nominated by students for the Excellence in Teaching Award (both years).

Scholarships and Honours

2019–2025 Fully funded doctoral scholarship, Imperial Business School (Graduate Teaching Assistant Scholarship)

2014–2016 University Honors, University of Michigan (three terms)

Skills and Languages

Programming	Python (NumPy, pandas, SciPy, scikit-learn, statsmodels, PyTorch), MATLAB, R, SQL, Git
Methods	Cross-sectional asset pricing (IPCA, QZ-IPCA, NA-IPCA, Fama–MacBeth), latent-factor models, regularised regression and portfolio shrinkage (ridge-type rules, UPSA), robust PCA (RPCA, SPCP), moving-block bootstrap inference, Monte Carlo simulation design, neural networks and transfer learning, state-space and regime-switching models, financial text analysis
Data	CRSP/Compustat and the Jensen–Kelly–Pedersen characteristic library (jkpfactors.com) via WRDS; Bloomberg; LSEG Workspace
Languages	Chinese (Mandarin, native); English (fluent; working language)

References

Available on request.

Updated September 2026